

High-Level-Synthesis-Flow-on-Zynq-using-Vivado-HLS

This course provides users with an understanding of high-level synthesis design methodologies necessary to develop digital systems using Vivado HLS 2018.2 version.

Required Setup

Clone this repository and then copy the **sources** directory as **labs** to a working directory. The recommended working directory is **c:/xup/hls**. After copying the directory, all labs will be carried out in their respective folders located at **c:/xup/hls/labs**.

In the instructions for the labs;

{labs} refers to **c:/xup/hls/labs**.

NOTE

Board support for the PYNQ-Z1/Z2 are not included in Vivado 2018.2 by default. The relevant files need to be extracted and saved to:

{Vivado installation}\data\boards\board_files\zynq

These files can be downloaded from

PYNQ-Z1:[/board files](#).

PYNQ-Z2:[/board files](#).

Hardware Setup

PYNQ-Z1/Z2: Connect the board to the PC using a micro USB cable. Make sure that a jumper is connected to JTAG (between JP1_1 and JP1_2) to use the board in the development mode. Also, make sure that another jumper is placed between J9_2 and J9_3 to select USB as a power source.

Labs Overview

Lab1:

This lab provides a basic introduction to high-level synthesis using the Vivado HLS tool flow. You will use

Vivado HLS in GUI mode to create a project. You will simulate, synthesize, perform design analysis, and implement the provided model.

Lab2:

This lab introduces various techniques and directives which can be used in Vivado HLS to improve design performance. The design under consideration accepts an image in a (custom) RGB format, converts it to the Y'UV color space, applies a filter to the Y'UV image and converts it back to RGB.

Lab3:

This lab introduces various techniques and directives which can be used in Vivado HLS to improve design performance as well as area and resource utilization. The design under consideration performs discrete cosine transformation (DCT) on an 8x8 block of data.

Lab4:

This lab introduces a design flow to generate a IP-XACT adapter from a design using Vivado HLS and using the generated IP-XACT adapter in a processor system using IP Integrator in Vivado.

Note. Since PYNQ-Z1 board does not have Audio CODEC chip, this lab cannot be ported/verified on the board, though design flow is still applicable to the board.